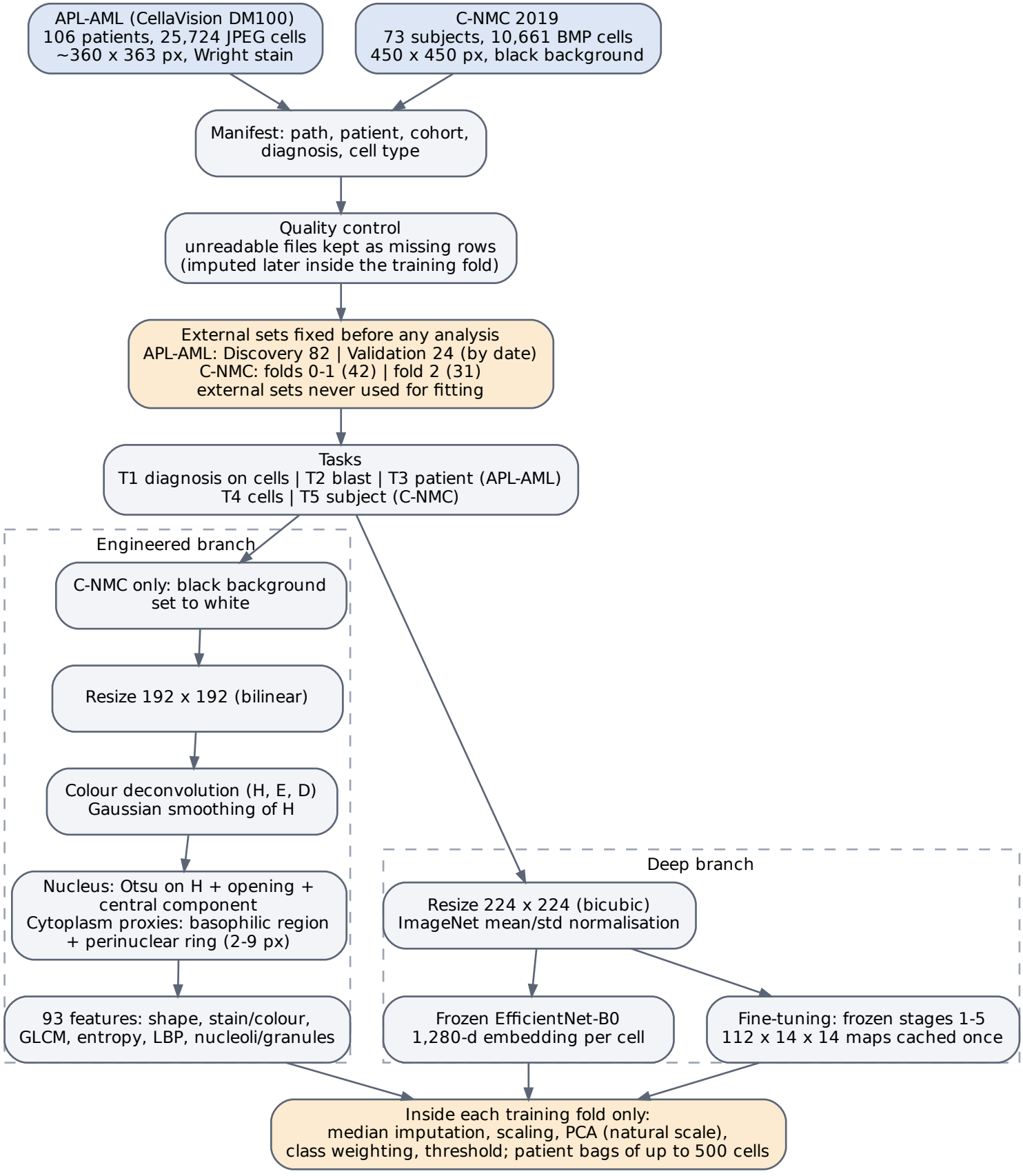
APL-AML (CellaVision DM100)
106 patients, 25,724 JPEG cells
~360 x 363 px, Wright stain
C-NMC 2019
73 subjects, 10,661 BMP cells
450 x 450 px, black background
Manifest: path, patient, cohort, diagnosis, cell type
Quality control
unreadable files kept as missing rows
(imputed later inside the training fold)
External sets fixed before any analysis
APL-AML: Discovery 82 | Validation 24 (by date)
C-NMC: folds 0-1 (42) | fold 2 (31)
external sets never used for fitting
Tasks
T1 diagnosis on cells | T2 blast | T3 patient (APL-AML)
T4 cells | T5 subject (C-NMC)
Engineered branch
C-NMC only: black background set to white
Resize 192 x 192 (bilinear)
Colour deconvolution (H, E, D)
Gaussian smoothing of H
Nucleus: Otsu on H + opening + central component
Cytoplasm proxies: basophilic region + perinuclear ring (2-9 px)
93 features: shape, stain/colour, GLCM, entropy, LBP, nucleoli/granules
Deep branch
Resize 224 x 224 (bicubic)
ImageNet mean/std normalisation
Frozen EfficientNet-B0
1,280-d embedding per cell
Fine-tuning: frozen stages 1-5
112 x 14 x 14 maps cached once
Inside each training fold only:
median imputation, scaling, PCA (natural scale),
class weighting, threshold; patient bags of up to 500 cells